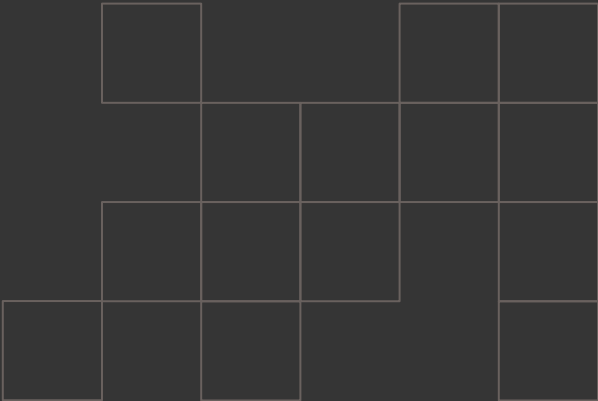


# PrintIT

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# When efficiency matters, tools need to be able to keep up.

Being able to print labels in an efficient way to keep up with shopping or simply to not have to spend too much time individually printing labels.



Who is this for?

## Businesses!

Any business which uses some sort of label and needs to print them regularly.



## What does PrintIT do?

1

### Organizes

Keeps all labels into different categories and makes it easy to stay organized.

2

### Queries

Quick searching cuts down on time lost when trying to find a certain label.

3

### Improves efficiency

No longer need to manually search through an entire file system to find labels

4

### Scalable

Able to easily add categories and labels so that when more products are added so can the labels.

## How is PrintIT used?

1

### Add Labels into database

Manually customize each entry into database to ensure PrintIT working properly.

2

### Search and print labels

Use program to search and print labels.

3

### Profit

Spend less time searching for a label and more time serving customers.

## SQLite Database

1

### Easy CRUD

Able to customize functions and sql queries to how they need to be used in the context of PrintIT

2

### Small size

Makes PrintIT lightweight and easy to use on any device connected to a printer.

3

### Offline capabilities

Continue to print even offline so that business operations do not halt during blackouts.

## SQLite database

```
public class Database { @Implicit +1*
    public static String url = "jdbc:sqlite:labels.db";

    public static void createDatabase() @Implicit
    {...}

    public static void checkDatabase() @Implicit
    {...}

    public static void createTable(String table) @Quentin Bordelon +1
    {...}

    public static ItemLabel genericGet(String fieldSearching, String table, String labelField) {...}

    public static ItemLabel getByID(Integer labelID, String table) {...}

    public static void addToDatabase(ItemLabel label, String table) @Implicit
    {...}

    public static void upsertToDatabase(int id, ItemLabel newLabel, String table) @Implicit
    {...}

    public static void deleteByID(int LabelID, String table) @Implicit
    {...}

    public static ArrayList<String> getAllTables() @Implicit
    {...}

    public static ArrayList<ItemLabel> getAllEntriesInTable(String table) throws SQLException {...}

    public static ArrayList<ItemLabel> searchForEntries(String table, String searchText) throws SQLException {...}

    public static ArrayList<ItemLabel> genericSearch(String searchText) throws SQLException @Implicit +1
    {...}

    public static ArrayList<ItemLabel> searchAllTables() throws SQLException @Implicit +1
    {...}

    public static ResultSet executeQuery(String query) @Implicit
    {...}

    public static ArrayList<ItemLabel> getAndTransform(ResultSet rs) throws SQLException {...}
}
```

## Printer Service

1

### Error free printing

Paper source and destination is determined based on the label type.

2

### More efficient printing

Less time spent in print pop up windows clicking.



```
public class Printing 3 usages ⓘ Implicitly
{

    public static Queue<ItemLabel> printerQueue = new ArrayDeque<>(); 3 usages
    public static Printer defaultPrinter = Printer.getDefaultPrinter(); 2 usages
    public static Set<PaperSource> supportedSources = getPrinterSources(); 1 usage
    public static PaperSource defaultSource = supportedSources.iterator().next(); 1 usage

    public static void addToQueue(ItemLabel itemLabel) { printerQueue.add(itemLabel); }

    public static void startPrintQueue() throws PrinterException, IOException 1 usage ⓘ Implicitly
    {...}

    public static void printSingle(String location) throws IOException, PrinterException {...}

    public static PrintService[] getPrintersAvailable() { return PrintServiceLookup.lookupPrintServices( flavor: null, attributes: null); }

    public static void setPrinter(Printer printer) { defaultPrinter = printer; }

    public static Set<PaperSource> getPrinterSources() 1 usage ⓘ Implicitly
    {...}

    public static void setPrinterSource(PaperSource newSource) { defaultSource = newSource; }

}
```

## Print service

1

### Easy to expand

Simple HTML like syntax which makes building UI quick and simple.

2

### Simple styling

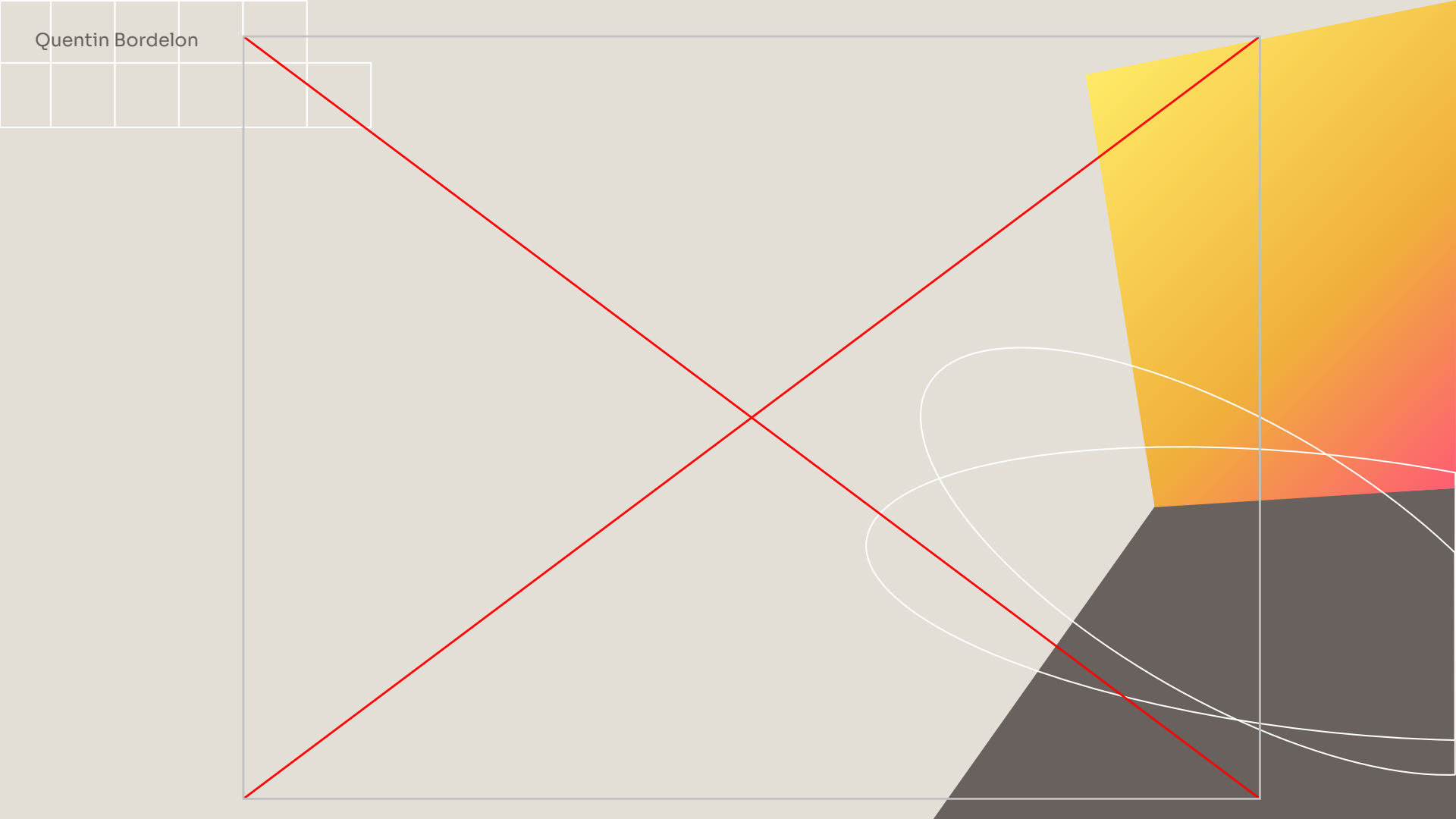
Same styling as a regular webpage making it simple to style JavaFX programs.

3

### Readable/maintainable code

FXML style makes views very easy to read. Maintainable format allowing for quick iteration and development

```
<VBox alignment="CENTER" prefHeight="400.0" prefWidth="800.0" spacing="20.0" xmlns:fx="http://javafx.com/fxml/1" xmlns="http://javafx.com/javafx/25" fx:controller="com.implycitt.printit.Controllers.MainController">
<children>
<AnchorPane maxHeight="-Infinity" maxWidth="-Infinity" prefHeight="-1.0" prefWidth="-1.0" VBox.vgrow="ALWAYS">
<children>
<AnchorPane maxHeight="-Infinity" maxWidth="-Infinity" prefHeight="-1.0" prefWidth="-1.0" VBox.vgrow="ALWAYS">
<children>
<SplitPane dividerPositions="0.789277658628764" prefHeight="600.0" prefWidth="800.0">
<items>
<AnchorPane minHeight="0.0" minWidth="0.0">
<children>
<SplitPane dividerPositions="0.25882352941176473" prefHeight="600.0" prefWidth="626.0">
<items>
<AnchorPane minHeight="0.0" minWidth="0.0" prefHeight="160.0" prefWidth="180.6">
<children>
<ListView fx:id="categoryList" layoutX="0.0" layoutY="13.0" prefHeight="572.0" prefWidth="142.0" />
</children>
</AnchorPane>
<AnchorPane minHeight="0.0" minWidth="0.0" prefHeight="160.0" prefWidth="190.6">
<children>
<VBox fx:id="labelList" layoutX="11.0" layoutY="46.0" prefHeight="539.0" prefWidth="435.0" />
<TextField fx:id="searchBar" layoutX="11.0" layoutY="14.0" prefHeight="25.0" prefWidth="436.0" promptText="Search for a label..." />
</children>
</AnchorPane>
</items>
</SplitPane>
</children></AnchorPane>
<AnchorPane minHeight="0.0" minWidth="0.0">
<VBox spacing="5">
<children>
<TabPane layoutX="7.0" layoutY="6.0" maxHeight="1.7976931348623157E308" prefHeight="69.0" prefWidth="195.0" tabClosingPolicy="UNAVAILABLE">
<tab fx:id="labelTab" text="Label">
<AnchorPane minHeight="0.0" minWidth="0.0" prefHeight="180.0" prefWidth="200.0" />
</tab>
<tab fx:id="categoryTab" text="Category">
<AnchorPane minHeight="0.0" minWidth="0.0" prefHeight="180.0" prefWidth="200.0" />
</tab>
</TabPane>
<AnchorPane fx:id="rightSideBar" />
</children>
</VBox>
</AnchorPane>
</items>
</SplitPane>
</children>
</AnchorPane>
</children>
</AnchorPane>
</children>
<stylesheets>
<URL value="@../style/base.css" />
<URL value="@../style/main.css" />
</stylesheets>
</VBox>
</children>
</FXML>
```



## Conclusion

1

### Styling

Adding more styling to make the overall UI more appealing and user friendly

2

### Bug fixes

Fixes to responsive UI and database query issues.



# Thank you

<https://github.com/Implycitt/PrintIt>